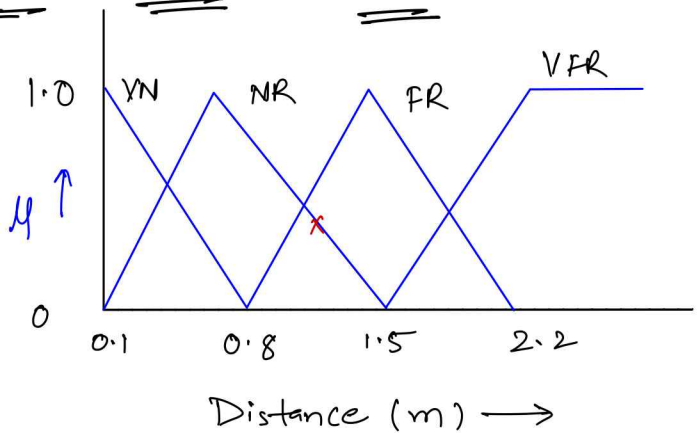
Navigation of Mobile Robot

Given Inputs: Distance = 1.04 m

Angle = 30°

For Distance = 1.04m



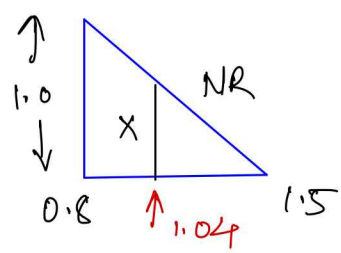
Step 1: Fuzzification.

⇒ Distance = 1.04 m is in between NR and FR

⇒ Therefore it may be either called NR (near) or FR (Far)

⇒ The membership values corresponding to μ_{NR} and μ_{FR} can be calculated from similar triangle rule.

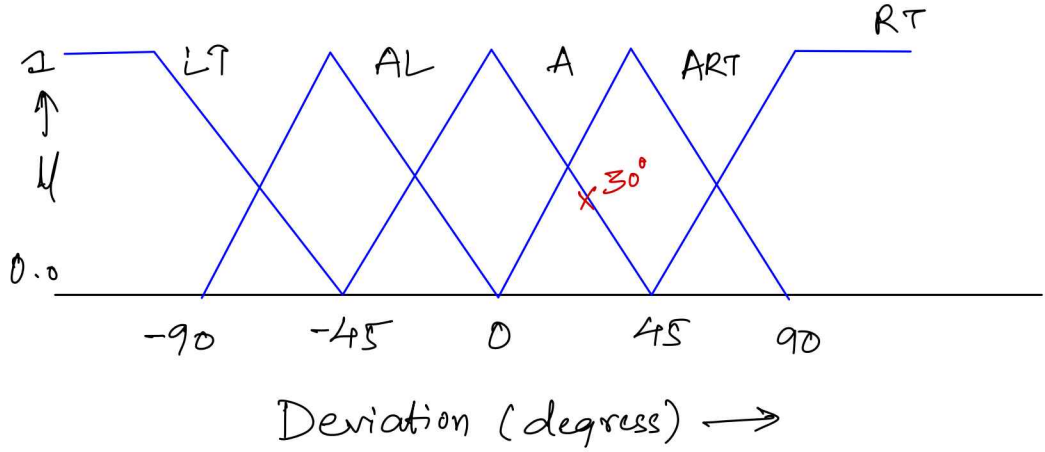
$$\boxed{\frac{x}{1.0} = \frac{1.5 - 1.04}{1.5 - 0.8}}$$



$\Rightarrow \quad \mu_{NR} = x = 0.6571$

$\Rightarrow \quad \mu_{FR} = 1 - \mu_{NR} = 0.3333.$

For Angle 30°



⇒ Similarly input angle of 30° may be A or ART

⇒ From similar triangle rule.

$\mu_A = 0.3333$

$\mu_{ART} = 0.6667$

Step 2: Inference Engine to find fired rules.

Out of 20 rules, only 4 rules are fired.

If Distance is NR AND Angle is A Then Deviation is RT

If Distance is NR AND Angle is ART Then Deviation is A

If Distance is FR AND Angle is A Then Deviation is ART

If Distance is FR AND Angle is ART Then Deviation is A

The strengths (α values) of fired rules are :

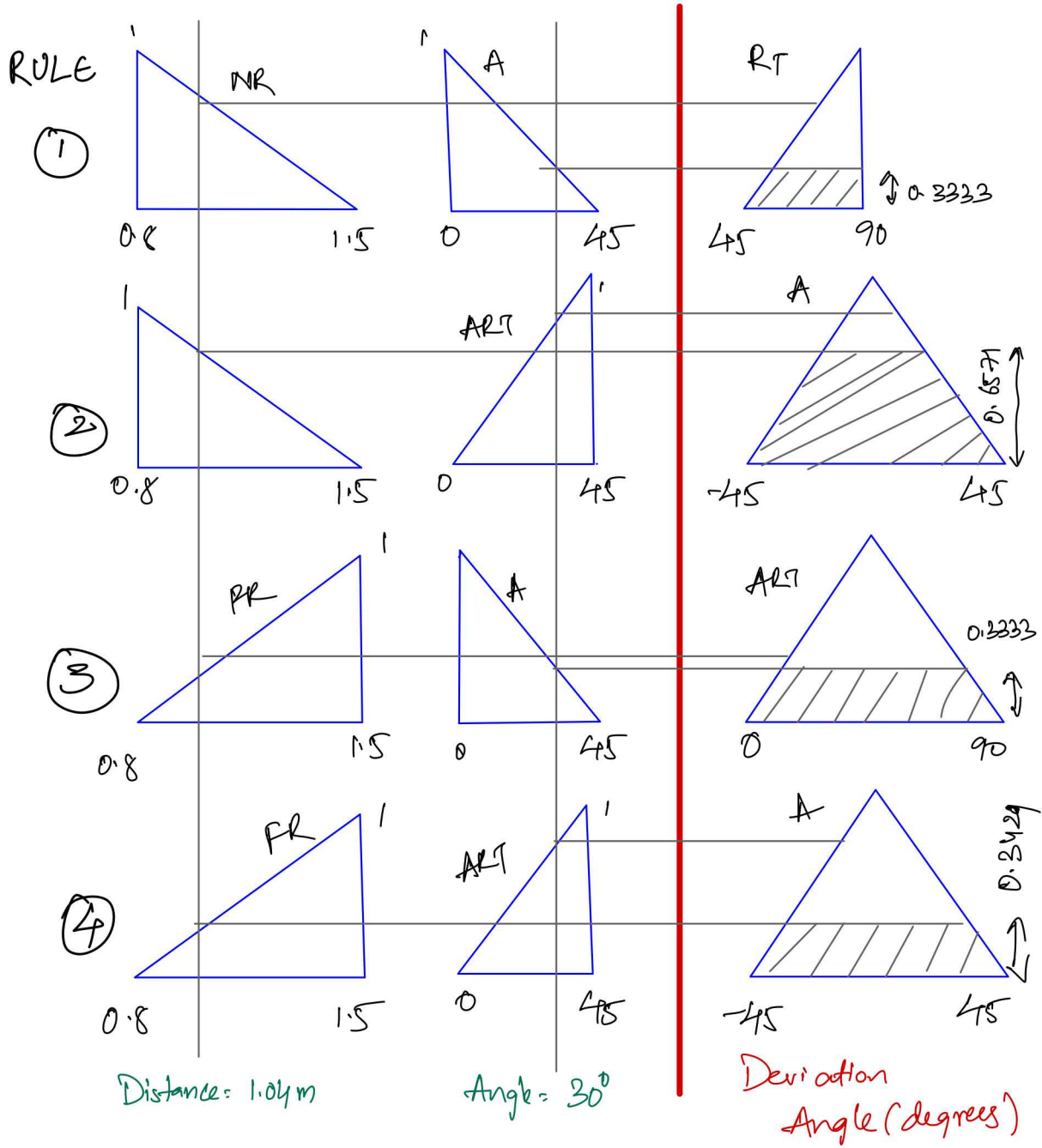
$\alpha_1 = \min(\mu_{NR}, \mu_A) = 0.3333$

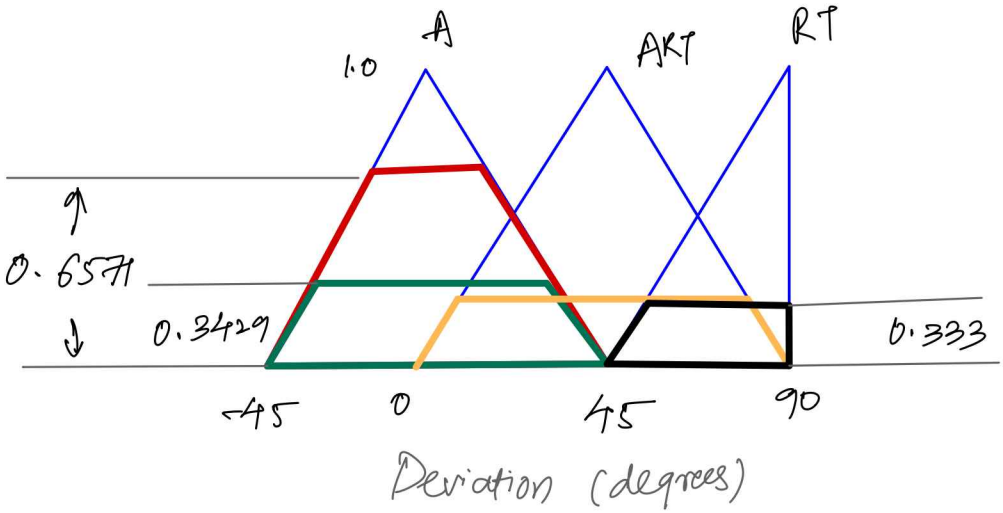
$\alpha_2 = \min(\mu_{NR}, \mu_{ART}) = 0.6571$

$\alpha_3 = \min(\mu_{FR}, \mu_A) = 0.3333$

$\alpha_4 = \min(\mu_{FR}, \mu_{ART}) = 0.3429$

The fuzzified Outputs corresponding to four different fired rules.





Combined fuzzified Output.

Step 3 : Defuzzification.

① Center of Sums method.

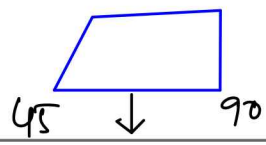
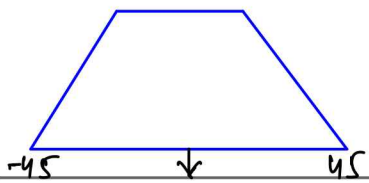
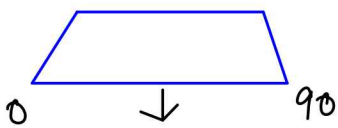
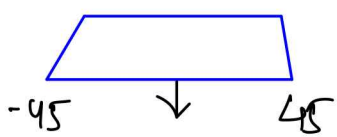
$$U'_{j'} = \frac{\sum_{j=1}^{p} A(\alpha_j) \times f_j}{\sum_{j=1}^{p} A(\alpha_j)}$$

$A(\alpha_j)$ → firing Area of jth rule

f_j → center of Area.

$U'_{j'}$ → Output of controller

Individualy calculate Area & center of area for each rule.

Fired Rule	Shaded Region	Area	Center
1		12.5	71
2		39.7019	0
3		25	45
4		25.5699	0

Area of Trapezoid

$$A = \frac{1}{2}(B_1 + B_2) \times H$$

B_1 → Base 1, B_2 → Base 2, H → Height

Centroid

$$C = \frac{B_1 + 2B_2}{3(B_1 + B_2)} \times H$$

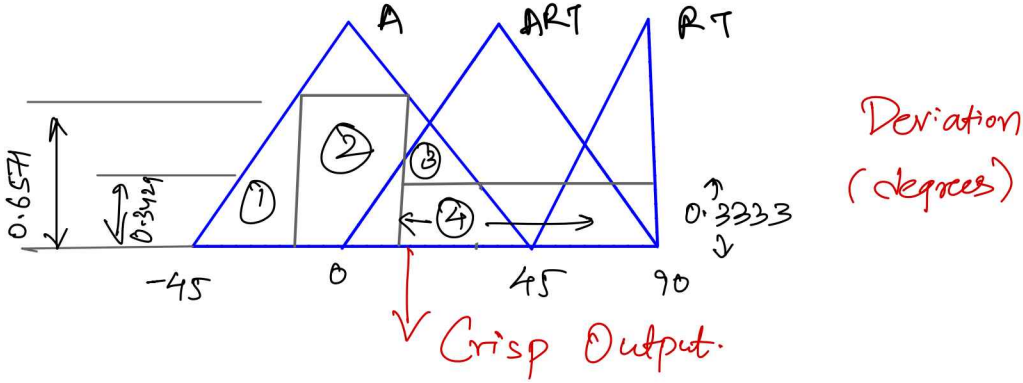
$$U = \frac{(12.5 \times 71) + (39.7089 \times 0) + (25 \times 45) + (25.6 \times 0)}{12.5 + 39.7089 + 25 + 25.5699}$$

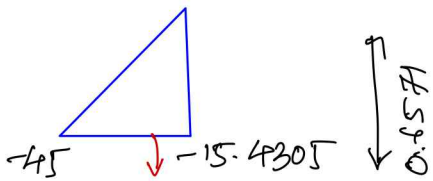
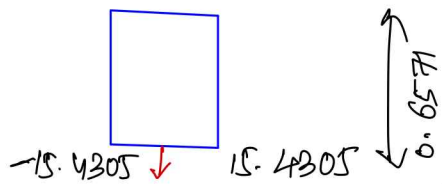
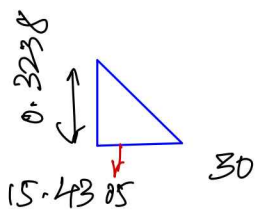
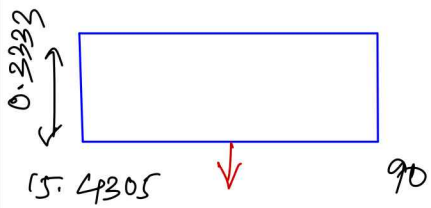
$$\boxed{U = 19.5809 \text{ degrees}} \rightarrow \text{Towards Right}$$

② Centroid Method

Centroid method is similar to center of sum method since it calculates Area and center of Area. (fast computing but less precise)

⇒ The key difference is instead of calculating Area & center for each output of a fired rule, we divide area and centre to known gemetry as simple triangle & rectangle and obtain output.



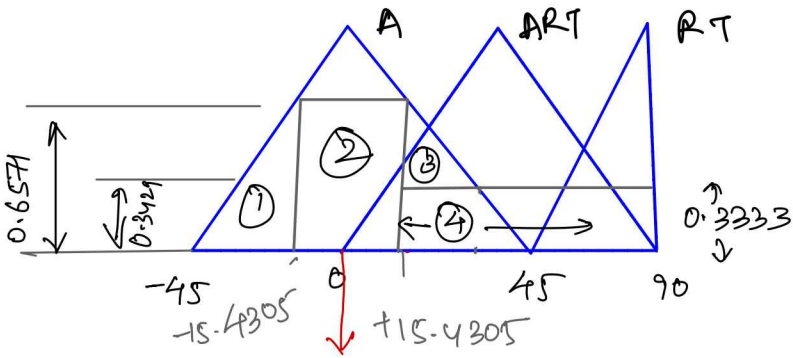
Marked Region	Shaded Region	Area	Center of Area
1		9.71	-25.28
2		20.27	0
3		2.35	20.28
4		24.85	52.71

$$U = \frac{(9.71 \times -25.28) + (20.27 \times 0) + (2.35 \times 20.28) + (24.85 \times 52.71)}{9.71 + 20.27 + 2.35 + 24.85}$$

$$\boxed{U = 19.4450}$$ → towards right.

③ Mean of Maxima Method.

⇒ Used when Output value is preferred as a crisp output (Integer)



Deviation (degrees)

⇒ Maximum value of membership is considered i.e. Area 2.

⇒ As the name suggests we need to calculate Mean of maximum membership value.

⇒ Range of Area = $\{-15.4305, 15.4305\}$

$$U = \text{Mean} = -15.4305 + 15.4305$$

$$\boxed{U = 0}$$

① Center of Sums → Precise but Computational cost is high

② Centroid → Fast calculation

③ Mean of Maxima → Integer Crisp Output is needed (Yes | No)